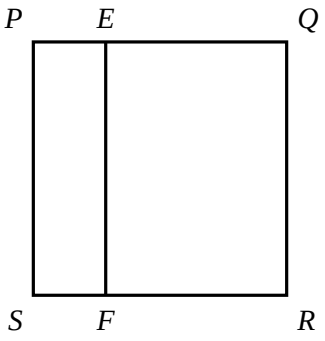
Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 42

Part A: Each question is worth 3 points

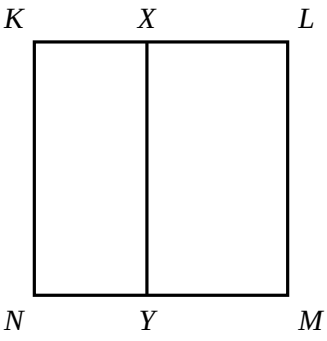
1. *PQRS* is a square. Its side is 7 cm. *PEFS* is a rectangle. Its shorter side is 2 cm.



How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 19 cm ***B.*** 18 cm ***C.*** 14 cm ***D.*** 10 cm ***E.*** 4 cm

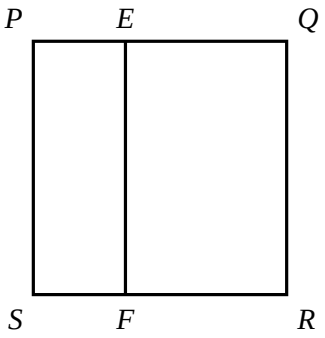
2. *KLMN* is a square. Its side is 9 cm. The line *XY* cuts it into two rectangles, and *KX* = 4 cm.



How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *XLMY*?

A. 5 cm ***B.*** 8 cm ***C.*** 10 cm ***D.*** 16 cm ***E.*** 28 cm

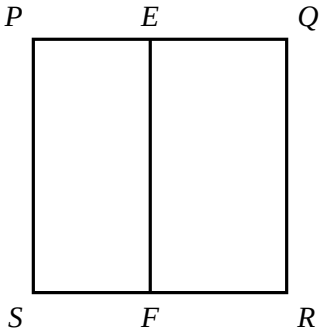
3. *PQRS* is a square. Its side is 11 cm. The line *EF* cuts it into two rectangles, and *PE* = 4 cm.



How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *EQRF*?

A. 8 cm ***B.*** 14 cm ***C.*** 16 cm ***D.*** 22 cm ***E.*** 36 cm

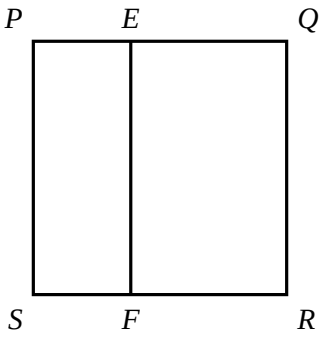
4. *PQRS* is a square. Its side is 13 cm. The line *EF* cuts it into two rectangles, and *PE* = 6 cm.



How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *EQRF*?

A. 24 cm ***B.*** 14 cm ***C.*** 12 cm ***D.*** 7 cm ***E.*** 6 cm

5. *PQRS* is a square. Its side is 13 cm. *PEFS* is a rectangle. Its shorter side is 5 cm.

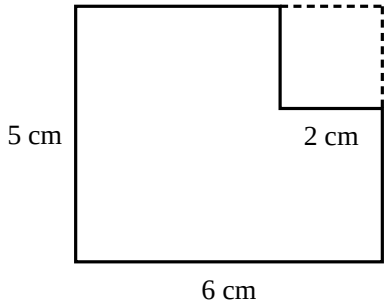


How many centimetres is the perimeter of the square *PQRS* larger than that of the rectangle *PEFS*?

A. 32 cm ***B.*** 26 cm ***C.*** 16 cm ***D.*** 10 cm ***E.*** 8 cm

Part B: Each question is worth 4 points

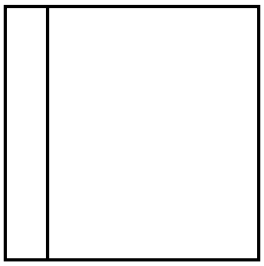
6. A rectangle is 6 cm long and 5 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 22 cm ***B.*** 20 cm ***C.*** 18 cm ***D.*** 14 cm ***E.*** 11 cm

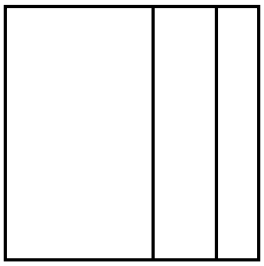
7. A square garden is surrounded by a fence 36 m long. The gardener divides it into 2 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 36 m ***B.*** 45 m ***C.*** 47 m ***D.*** 52 m ***E.*** 54 m

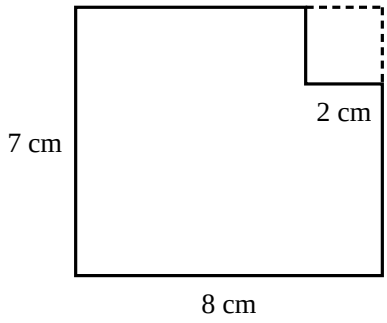
8. A square has perimeter 36 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 72 cm ***B.*** 78 cm ***C.*** 80 cm ***D.*** 90 cm ***E.*** 108 cm

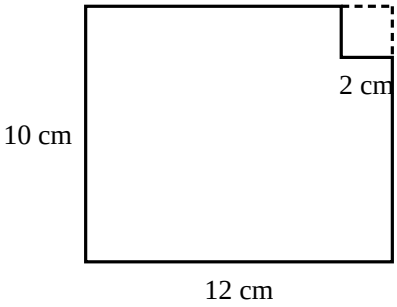
9. A rectangle is 8 cm long and 7 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 38 cm ***B.*** 34 cm ***C.*** 30 cm ***D.*** 26 cm ***E.*** 15 cm

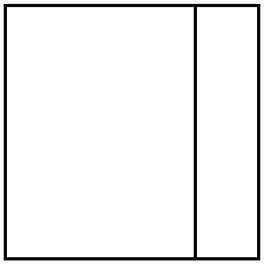
10. A rectangle is 12 cm long and 10 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 53 cm ***B.*** 52 cm ***C.*** 48 cm ***D.*** 47 cm ***E.*** 44 cm

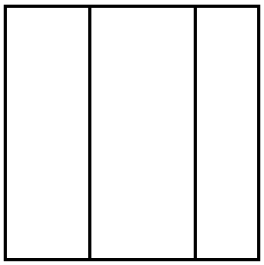
11. A square has perimeter 28 cm. It is cut along straight lines into 2 rectangles as shown.



What is the sum of the perimeters of all 2 rectangles?

A. 42 cm ***B.*** 43 cm ***C.*** 44 cm ***D.*** 51 cm ***E.*** 56 cm

12. A square garden is surrounded by a fence 48 m long. The gardener divides it into 3 rectangular beds as shown and puts a separate fence around each bed.

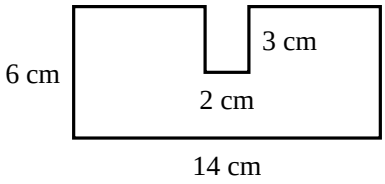


How many metres of fence are around the beds altogether?

A. 96 m ***B.*** 114 m ***C.*** 115 m ***D.*** 120 m ***E.*** 144 m

Part C: Each question is worth 5 points

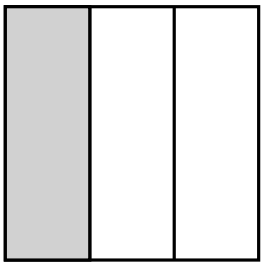
13. A 14 cm by 6 cm rectangle of cardboard has a rectangular notch 2 cm wide and 3 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 38 cm ***B.*** 44 cm ***C.*** 46 cm ***D.*** 48 cm ***E.*** 50 cm

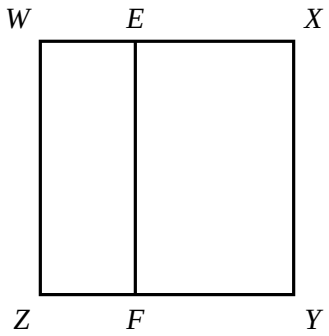
14. Mia cuts a square sheet of paper into 3 equal strips as shown. The perimeter of each strip is 48 cm.



What is the perimeter of the square?

A. 144 cm ***B.*** 96 cm ***C.*** 76 cm ***D.*** 72 cm ***E.*** 18 cm

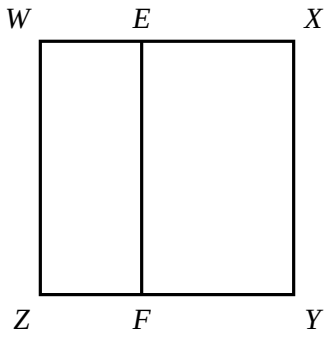
15. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 22 cm and the perimeter of *EXYF* is 26 cm.



How long is the side of the square *WXYZ*?

A. 8 cm ***B.*** 16 cm ***C.*** 32 cm ***D.*** 40 cm ***E.*** 48 cm

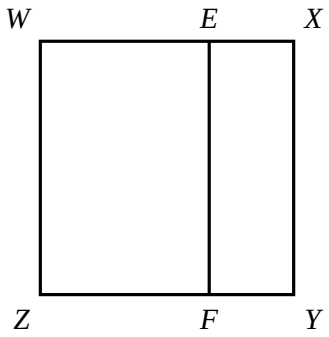
16. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 14 cm and the perimeter of *EXYF* is 16 cm.



How long is the side of the square *WXYZ*?

A. 1 cm ***B.*** 2 cm ***C.*** 3 cm ***D.*** 4 cm ***E.*** 5 cm

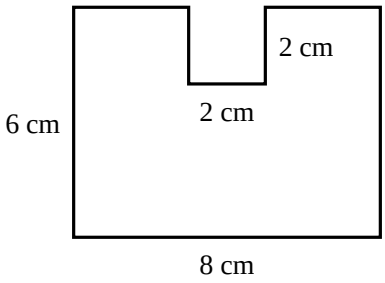
17. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 30 cm and the perimeter of *EXYF* is 24 cm.



How long is the side of the square *WXYZ*?

A. 9 cm ***B.*** 13 cm ***C.*** 18 cm ***D.*** 36 cm ***E.*** 45 cm

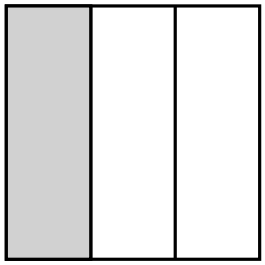
18. A 8 cm by 6 cm rectangle of cardboard has a rectangular notch 2 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 32 cm ***B.*** 36 cm ***C.*** 37 cm ***D.*** 40 cm ***E.*** 41 cm

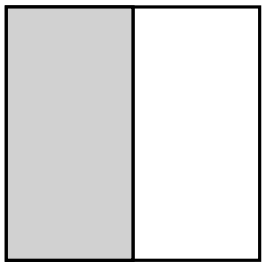
19. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 24 cm.



What is the perimeter of the square?

A. 72 cm ***B.*** 48 cm ***C.*** 36 cm ***D.*** 30 cm ***E.*** 9 cm

20. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 12 cm.



What is the perimeter of the square?

A. 16 cm ***B.*** 17 cm ***C.*** 18 cm ***D.*** 20 cm ***E.*** 24 cm

Answer Key

1.	**D**	**2.**	**B**	**3.**	**A**	**4.**	**C**	**5.**	**C**	**6.**	**A**
7.	**E**	**8.**	**A**	**9.**	**C**	**10.**	**E**	**11.**	**A**	**12.**	**A**
13.	**C**	**14.**	**D**	**15.**	**A**	**16.**	**E**	**17.**	**A**	**18.**	**A**
19.	**C**	**20.**	**A**								

Solutions

1. D (10 cm) — Square: 4 × 7 = 28 cm. Rectangle: 2 + 7 + 2 + 7 = 18 cm. Difference 28 – 18 = 10 cm. [2002 Q6 · square_minus_strip]

2. B (8 cm) — *XLMY* has sides 5 and 9 cm, perimeter 28 cm. Square: 36 cm. Difference 8 cm (= 2 × 4). [2002 Q6 · square_minus_strip]

3. A (8 cm) — *EQRF* has sides 7 and 11 cm, perimeter 36 cm. Square: 44 cm. Difference 8 cm (= 2 × 4). [2002 Q6 · square_minus_strip]

4. C (12 cm) — *EQRF* has sides 7 and 13 cm, perimeter 40 cm. Square: 52 cm. Difference 12 cm (= 2 × 6). [2002 Q6 · square_minus_strip]

5. C (16 cm) — Square: 4 × 13 = 52 cm. Rectangle: 5 + 13 + 5 + 13 = 36 cm. Difference 52 – 36 = 16 cm. [2002 Q6 · square_minus_strip]

6. A (22 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (6 + 5) = 22 cm. [2002 Q6 · corner_cut]

7. E (54 m) — Side = 36 ÷ 4 = 9 m. There is 1 cut; each cut is 9 m long and is counted twice (once for each piece next to it): 36 + 1 × 2 × 9 = 54 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

8. A (72 cm) — Side = 36 ÷ 4 = 9 cm. There are 2 cuts; each cut is 9 cm long and is counted twice (once for each piece next to it): 36 + 2 × 2 × 9 = 72 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

9. C (30 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (8 + 7) = 30 cm. [2002 Q6 · corner_cut]

10. E (44 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (12 + 10) = 44 cm. [2002 Q6 · corner_cut]

11. A (42 cm) — Side = 28 ÷ 4 = 7 cm. There is 1 cut; each cut is 7 cm long and is counted twice (once for each piece next to it): 28 + 1 × 2 × 7 = 42 cm. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

12. A (96 m) — Side = 48 ÷ 4 = 12 m. There are 2 cuts; each cut is 12 m long and is counted twice (once for each piece next to it): 48 + 2 × 2 × 12 = 96 m. The widths of the pieces don't matter! [2002 Q6 · pieces_total]

13. C (46 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 3 cm. Perimeter = 2 × (14 + 6) + 2 × 3 = 46 cm. [2002 Q6 · notch]

14. D (72 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 48 ÷ 8 = 6, so the side is 3 × 6 = 18 cm and the perimeter is 72 cm. [2002 Q6 · strips_to_square]

15. A (8 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 22 + 26 = 48 cm = 6 sides. Side = 48 ÷ 6 = 8 cm. [2002 Q6 · two_perimeters]

16. E (5 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 14 + 16 = 30 cm = 6 sides. Side = 30 ÷ 6 = 5 cm. [2002 Q6 · two_perimeters]

17. A (9 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 30 + 24 = 54 cm = 6 sides. Side = 54 ÷ 6 = 9 cm. [2002 Q6 · two_perimeters]

18. A (32 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (8 + 6) + 2 × 2 = 32 cm. [2002 Q6 · notch]

19. C (36 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 24 ÷ 8 = 3, so the side is 3 × 3 = 9 cm and the perimeter is 36 cm. [2002 Q6 · strips_to_square]

20. A (16 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 12 ÷ 6 = 2, so the side is 2 × 2 = 4 cm and the perimeter is 16 cm. [2002 Q6 · strips_to_square]